SDE-SQL: Self-Driven Database Exploration for Accurate Text-to-SQL

1. Project Overview

SDE-SQL is a production-quality prototype inspired by the paper: *SDE-SQL: Enhancing Text-to-SQL Generation in Large Language Models via Self-Driven Exploration with SQL Probes*. Instead of the traditional Text-to-SQL pipeline (User Question -> LLM -> SQL), this project implements a dynamic, self-driven exploration mechanism where the LLM interacts with the database via "SQL probes" before generating the final SQL query. It also iteratively refines queries if they produce errors or empty results.

2. Architecture & Pipeline

The SDE-SQL pipeline operates in several phases to ensure high accuracy: - **Schema Linking (Phase 3):** Extracts entities and candidate values from the question to create a Linked Schema. - **Exploration Manager (Phase 5):** Conducts a two-stage exploration. It generates SQL probes based on the Linked Schema, executes them against the database, and categorizes them into successful and rejected combinations. - **SQL Generation:** Uses the LLM to generate an initial SQL query based on the question, linked schema, and exploration results. - **Refinement Engine (Phase 7):** Executes the generated SQL. If it fails or returns an empty set, it uses a refinement loop with the LLM to fix the query based on database feedback. - **Target Checking (Phase 8):** Validates the final query to ensure it aligns with the expected output structure and intent.

3. Evaluation Metrics

The project is evaluated using an automated test suite (`evaluate.py`) that executes queries against a test dataset and measures the following key metrics: - **Execution Accuracy (EX):** The percentage of predicted SQL queries that produce the exact same result set (order-agnostic) as the ground truth (Gold SQL). - **Valid SQL Rate:** The percentage of generated queries that successfully execute against the database without syntax or structural errors (returning either rows or an empty set). - **Refinement Usage & Success Rate:** Tracks how often the Refinement Engine is triggered (due to initial failures/empty results) and what percentage of those refinements successfully fix the query. - **Average Probes per Question:** The average number of SQL probes executed during the Two-Stage Exploration phase to gather database context. - **Average Latency:** The total execution time (in milliseconds) from user question to final refined SQL output, accounting for database round-trips and LLM inference. - **Average Refinement Iterations:** The mean number of refinement loops required to arrive at a valid query.

4. Baseline Evaluation

To demonstrate the effectiveness of the SDE-SQL approach, the system compares the full pipeline against a standard Baseline Pipeline. - **Baseline Pipeline:** A traditional zero-shot approach (Question -> LLM -> SQL) without schema linking, database exploration, or refinement. - **SDE-SQL Advantage:** 1. **Higher Accuracy:** By actively probing the database (e.g., checking if a specific column value exists), SDE-SQL avoids hallucinating data values. 2. **Self-Correction:** The Refinement Engine ensures that minor syntax errors or incorrect joins are caught and fixed before returning to the user, significantly boosting the Valid SQL Rate compared to the baseline. 3. **Context Awareness:** Schema linking restricts the LLM to relevant tables and columns, reducing the search space and minimizing hallucinations compared to providing the entire database schema as in the baseline.